Robotics





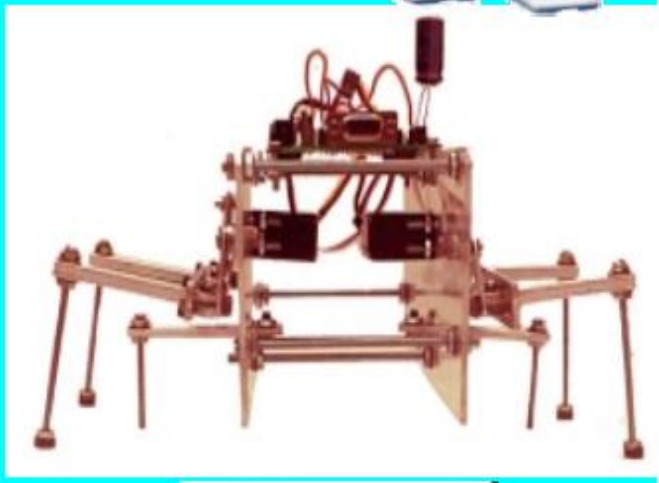




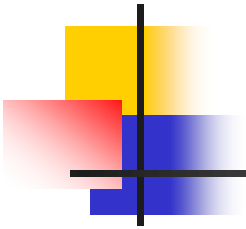
A255 robot









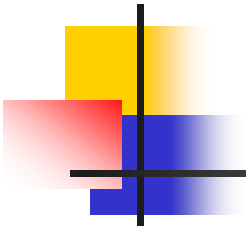


Motivation

- Intelligent Environments are aimed at improving the inhabitants' experience and task performance
  - Automate functions in the home
  - Provide services to the inhabitants
- Decisions coming from the decision maker(s) in the environment have to be executed.
  - Decisions require actions to be performed on devices
  - Decisions are frequently not elementary device interactions but rather relatively complex commands
    - Decisions define set points or results that have to be achieved
    - Decisions can require entire tasks to be performed

Automation and Robotics in Intelligent Environments

- Control of the physical environment
  - Automated blinds
  - Thermostats and heating ducts
  - Automatic doors
  - Automatic room partitioning
- Personal service robots
  - House cleaning
  - Lawn mowing
  - Assistance to the elderly and handicapped
  - Office assistants
  - Security services



Robots

- Robota (Czech) = A worker of forced labor

  From Czech playwright Karel Capek's 1921 play “R.U.R” (“Rossum's Universal Robots”)
- Japanese Industrial Robot Association (JIRA) :

  “A device with degrees of freedom that can be controlled.”
  - Class 1 : Manual handling device
  - Class 2 : Fixed sequence robot
  - Class 3 : Variable sequence robot
  - Class 4 : Playback robot
  - Class 5 : Numerical control robot
  - Class 6 : Intelligent robot

A Brief History of Robotics

- Mechanical Automata
  - Ancient Greece & Egypt
    - Water powered for ceremonies
  - 14th – 19th century Europe
    - Clockwork driven for entertainment
- Motor driven Robots
  - 1928: First motor driven automata
  - 1961: Unimate
    - First industrial robot
  - 1967: Shakey
    - Autonomous mobile research robot
  - 1969: Stanford Arm
    - Dextrous, electric motor driven robot arm



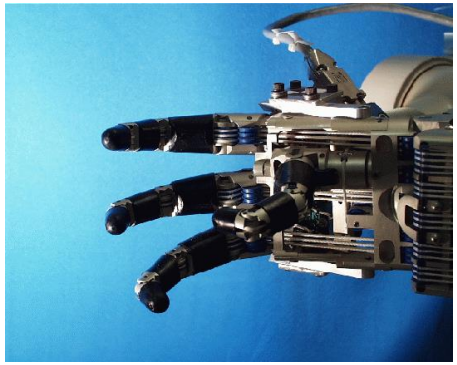
Maillardet's Automaton

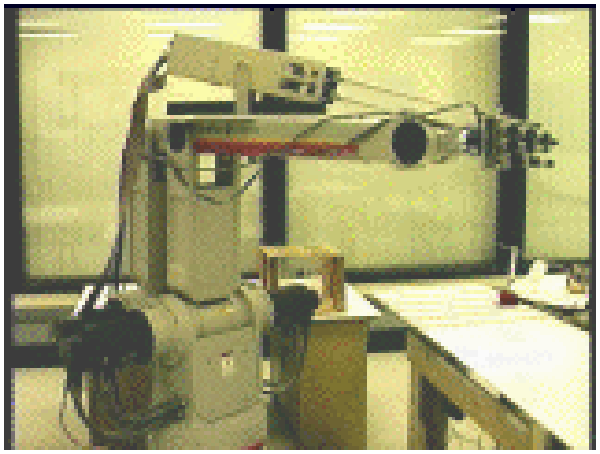


Unimate

Robots

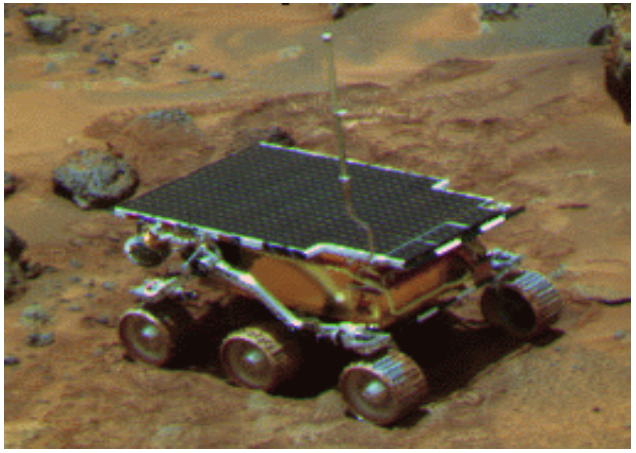
- Robot Manipulators





- Mobile Robots

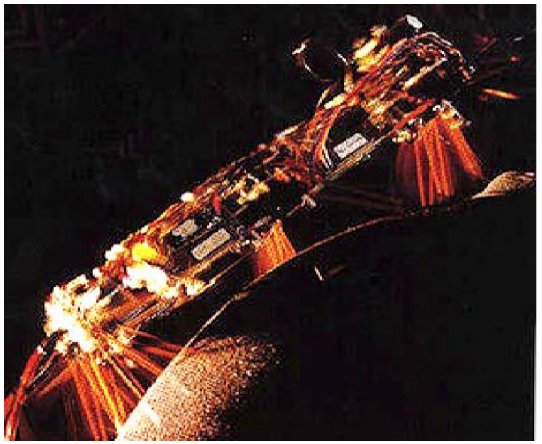




Robots

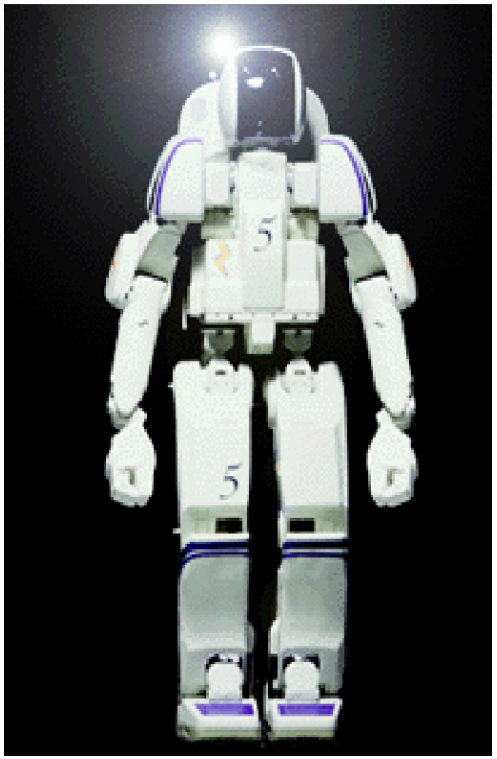
- Walking Robots

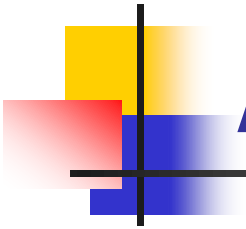




- Humanoid Robots







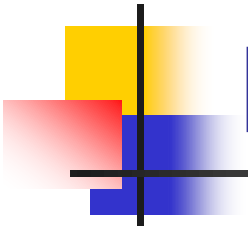
Autonomous Robots

- The control of autonomous robots involves a number of subtasks
  - Understanding and modeling of the mechanism
    - Kinematics, Dynamics, and Odometry
  - Reliable control of the actuators
    - Closed-loop control
  - Generation of task-specific motions
    - Path planning
  - Integration of sensors
    - Selection and interfacing of various types of sensors
  - Coping with noise and uncertainty
    - Filtering of sensor noise and actuator uncertainty
  - Creation of flexible control policies
    - Control has to deal with new situations

Traditional Industrial Robots

- Traditional industrial robot control uses robot arms and largely pre-computed motions
  - Programming using "teach box"
  - Repetitive tasks
  - High speed
  - Few sensing operations
  - High precision movements
  - Pre-planned trajectories and task policies
  - No interaction with humans

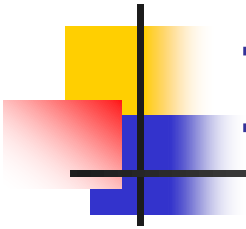




Problems

- Traditional programming techniques for industrial robots lack key capabilities necessary in intelligent environments
  - Only limited on-line sensing
  - No incorporation of uncertainty
  - No interaction with humans
  - Reliance on perfect task information
  - Complete re-programming for new tasks

Requirements for Robots in Intelligent Environments



- Autonomy
  - Robots have to be capable of achieving task objectives without human input
  - Robots have to be able to make and execute their own decisions based on sensor information
- Intuitive Human-Robot Interfaces
  - Use of robots in smart homes can not require extensive user training
  - Commands to robots should be natural for inhabitants
- Adaptation
  - Robots have to be able to adjust to changes in the environment

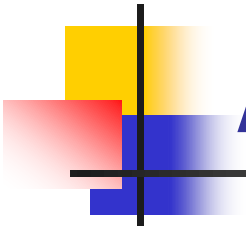
Robots for Intelligent Environments

- Service Robots
  - Security guard
  - Delivery
  - Cleaning
  - Mowing
- Assistance Robots
  - Mobility
  - Services for elderly and People with disabilities







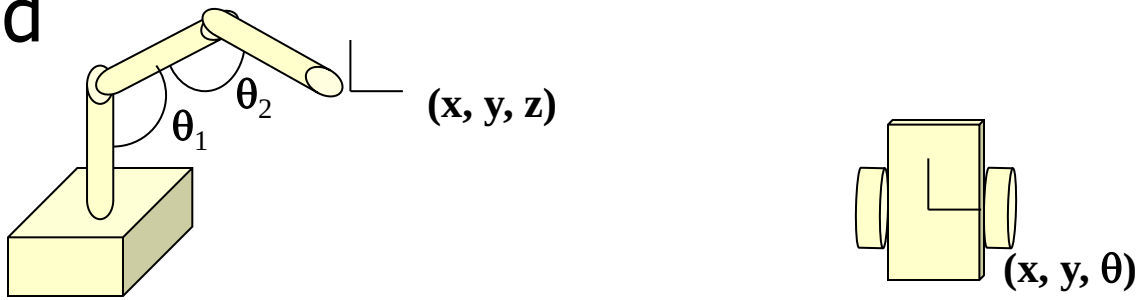


Autonomous Robot Control

- To control robots to perform tasks autonomously a number of tasks have to be addressed:
  - Modeling of robot mechanisms
    - Kinematics, Dynamics
  - Robot sensor selection
    - Active and passive proximity sensors
  - Low-level control of actuators
    - Closed-loop control
  - Control architectures
    - Traditional planning architectures
    - Behavior-based control architectures
    - Hybrid architectures

Modeling the Robot Mechanism

- Forward kinematics describes how the robots joint angle configurations translate to locations in the world



- Inverse kinematics computes the joint angle configuration necessary to reach a particular point in space.
- Jacobians calculate how the speed and configuration of the actuators translate into velocity of the robot

Mobile Robot Odometry

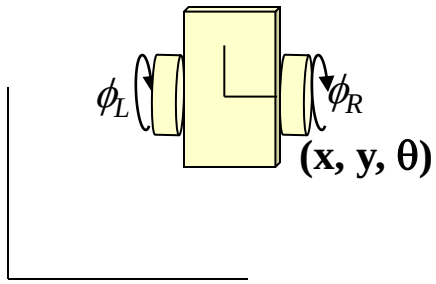
- In mobile robots the same configuration in terms of joint angles does not identify a unique location
  - To keep track of the robot it is necessary to incrementally update the location (this process is called odometry or dead reckoning)

$$\begin{pmatrix} x \\ y \\ \theta \end{pmatrix}^{t+\Delta t} = \begin{pmatrix} x \\ y \\ \theta \end{pmatrix}^{t} + \begin{pmatrix} v_x \\ v_y \\ \varpi \end{pmatrix} \Delta t$$

  - Example: A differential drive robot

$$v_x = \cos(\theta)\frac{r(\dot{\phi}_L + \dot{\phi}_R)}{2}, v_y = \sin(\theta)\frac{r(\dot{\phi}_L + \dot{\phi}_R)}{2}$$

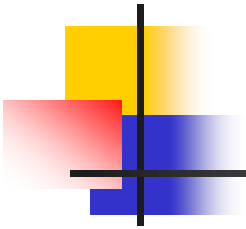
$$\varpi = \frac{r}{d}\left(\dot{\phi}_L - \dot{\phi}_R\right)$$



Actuator Control

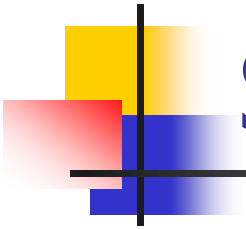
- To get a particular robot actuator to a particular location it is important to apply the correct amount of force or torque to it.
  - Requires knowledge of the dynamics of the robot
    - Mass, inertia, friction
    - For a simplistic mobile robot: $F = m\,a + B\,v$
  - Frequently actuators are treated as if they were independent (i.e. as if moving one joint would not affect any of the other joints).
  - The most common control approach is PD-control (proportional, differential control)
    - For the simplistic mobile robot moving in the x direction:

$$F = K_P\left(x_{desired} - x_{actual}\right) + K_D\left(v_{desired} - v_{actual}\right)$$



Robot Navigation

- Path planning addresses the task of computing a trajectory for the robot such that it reaches the desired goal without colliding with obstacles
  - Optimal paths are hard to compute in particular for robots that can not move in arbitrary directions (i.e. nonholonomic robots)
  - Shortest distance paths can be dangerous since they always graze obstacles
  - Paths for robot arms have to take into account the entire robot (not only the endeffector)



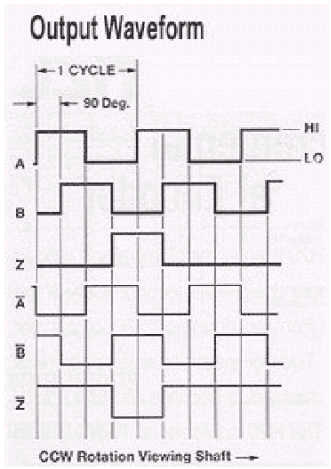
Sensor-Driven Robot Control

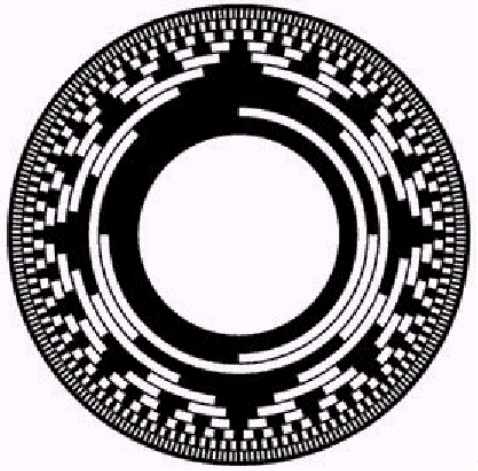
- To accurately achieve a task in an intelligent environment, a robot has to be able to react dynamically to changes ion its surrounding
  - Robots need sensors to perceive the environment
  - Most robots use a set of different sensors
    - Different sensors serve different purposes
  - Information from sensors has to be integrated into the control of the robot

Robot Sensors

- Internal sensors to measure the robot configuration
  - Encoders measure the rotation angle of a joint



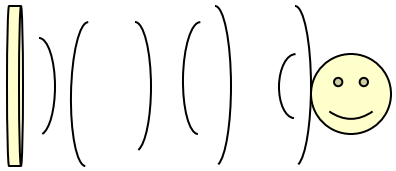




  - Limit switches detect when the joint has reached the limit

Robot Sensors

- Proximity sensors are used to measure the distance or location of objects in the environment. This can then be used to determine the location of the robot.
  - Infrared sensors determine the distance to an object by measuring the amount of infrared light the object reflects back to the robot
  - Ultrasonic sensors (sonars) measure the time that an ultrasonic signal takes until it returns to the robot





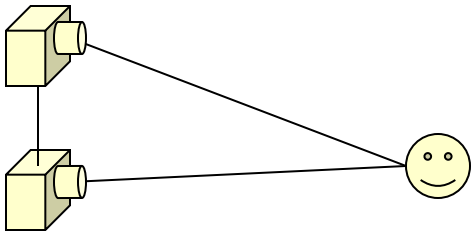
  - Laser range finders determine distance by measuring either the time it takes for a laser beam to be reflected back to the robot or by measuring where the laser hits the object



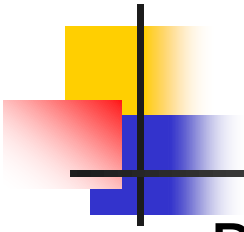
Robot Sensors

- Computer Vision provides robots with the capability to passively observe the environment
  - Stereo vision systems provide complete location information using triangulation





  - However, computer vision is very complex
    - Correspondence problem makes stereo vision even more difficult



Uncertainty in Robot Systems

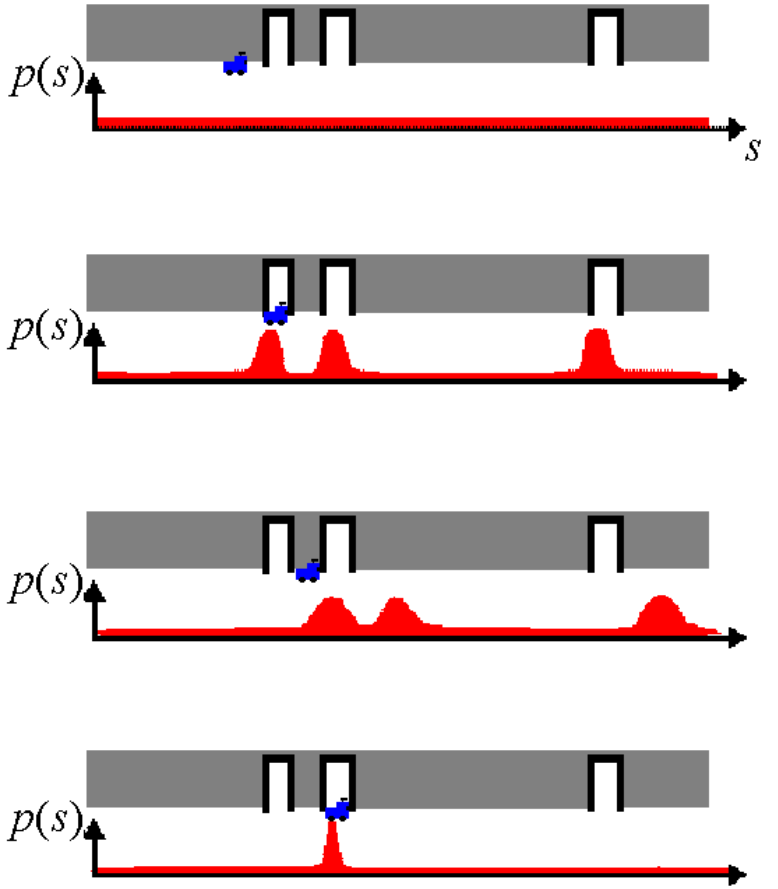
- Robot systems in intelligent environments have to deal with sensor noise and uncertainty
  - Sensor uncertainty
    - Sensor readings are imprecise and unreliable
  - Non-observability
    - Various aspects of the environment can not be observed
    - The environment is initially unknown
  - Action uncertainty
    - Actions can fail
    - Actions have nondeterministic outcomes

Probabilistic Robot Localization

- Explicit reasoning about Uncertainty using Bayes filters:

$$b(x_t) = \eta\ p(o_t \mid x_t) \int p(x_t \mid x_{t-1}, a_{t-1})\ b(x_{t-1})\ dx_{t-1}$$

- Used for:
  - Localization
  - Mapping
  - Model building



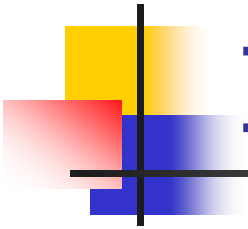
Traditional Human-Robot Interface: Teleoperation

- Remote Teleoperation: Direct operation of the robot by the user
  - User uses a 3-D joystick or an exoskeleton to drive the robot
    - Simple to install
    - Removes user from dangerous areas
  - Problems:
    - Requires insight into the mechanism
    - Can be exhaustive
    - Easily leads to operation errors





Human-Robot Interaction in Intelligent Environments



- Personal service robot
  - Controlled and used by untrained users
    - Intuitive, easy to use interface
    - Interface has to "filter" user input
      - Eliminate dangerous instructions
      - Find closest possible action
  - Receive only intermittent commands
    - Robot requires autonomous capabilities
    - User commands can be at various levels of complexity
    - Control system merges instructions and autonomous operation
  - Interact with a variety of humans
    - Humans have to feel "comfortable" around robots
    - Robots have to communicate intentions in a natural way

Example: Minerva the Tour Guide Robot (CMU/Bonn)

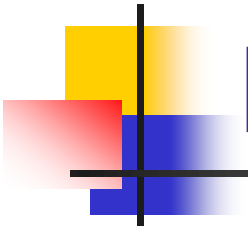


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http://www.cs.cmu.edu/~thrun/movies/minerva.mpg

Intuitive Robot Interfaces: Command Input

- Graphical programming interfaces
  - Users construct policies form elemental blocks
  - Problems:
    - Requires substantial understanding of the robot
- Deictic (pointing) interfaces
  - Humans point at desired targets in the world or
  - Target specification on a computer screen
  - Problems:
    - How to interpret human gestures ?
- Voice recognition
  - Humans instruct the robot verbally
  - Problems:
    - Speech recognition is very difficult
    - Robot actions corresponding to words has to be defined

Intuitive Robot Interfaces: Robot-Human Interaction

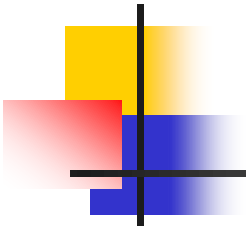


- The robot has to be able to communicate its intentions to the human
  - Output has to be easy to understand by humans
  - Robot has to be able to encode its intention
  - Interface has to keep human's attention without annoying her
- Robot communication devices:
  - Easy to understand computer screens
  - Speech synthesis
  - Robot "gestures"

Example: The Nursebot Project



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http://www/cs/cmu.edu/~thrun/movies/pearl_assist.mpg



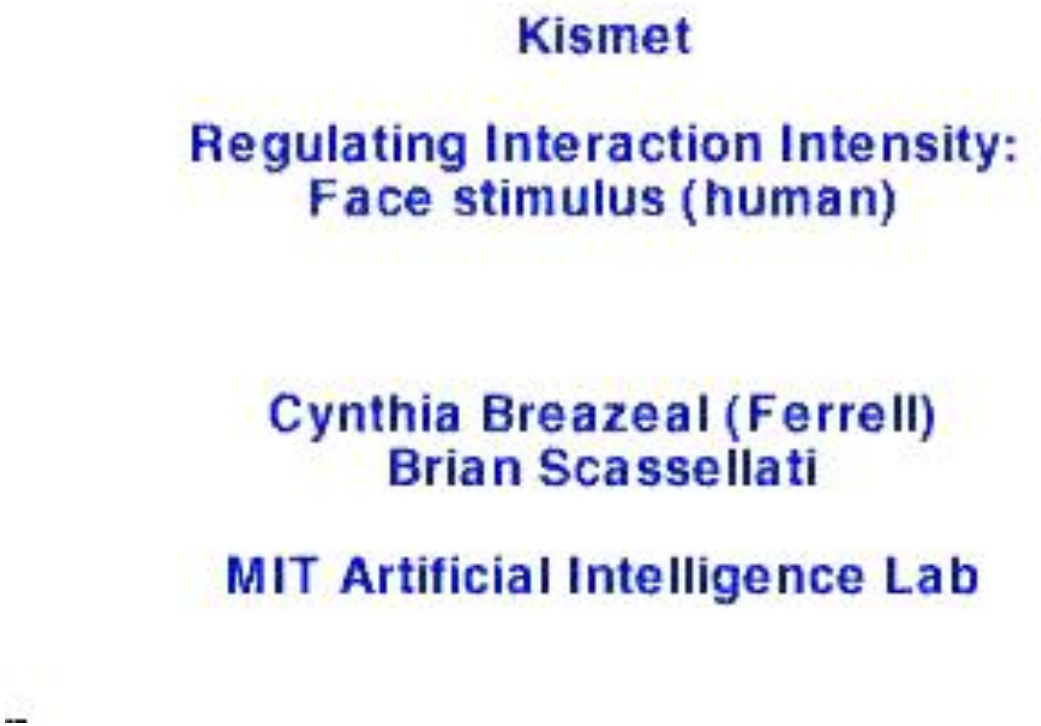
Human-Robot Interfaces

- Existing technologies
  - Simple voice recognition and speech synthesis
  - Gesture recognition systems
  - On-screen, text-based interaction
- Research challenges
  - How to convey robot intentions ?
  - How to infer user intent from visual observation (how can a robot imitate a human) ?
  - How to keep the attention of a human on the robot ?
  - How to integrate human input with autonomous operation ?

"Social" Robot Interactions

- To make robots acceptable to average users they should appear and behave “natural”
  - "Attentional" Robots
    - Robot focuses on the user or the task
    - Attention forms the first step to imitation
  - "Emotional" Robots
    - Robot exhibits “emotional” responses
    - Robot follows human social norms for behavior
      - Better acceptance by the user (users are more forgiving)
      - Human-machine interaction appears more “natural”
      - Robot can influence how the human reacts

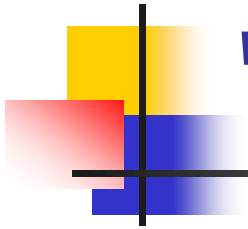
"Social" Robot Example: Kismet



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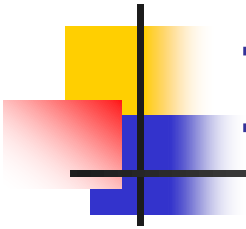
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"Social" Robot Interactions



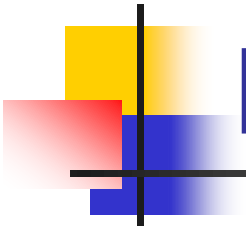
- Advantages:
  - Robots that look human and that show “emotions” can make interactions more “natural”
    - Humans tend to focus more attention on people than on objects
    - Humans tend to be more forgiving when a mistake is made if it looks “human”
  - Robots showing “emotions” can modify the way in which humans interact with them
- Problems:
  - How can robots determine the right emotion ?
  - How can “emotions” be expressed by a robot ?

Human-Robot Interfaces for Intelligent Environments



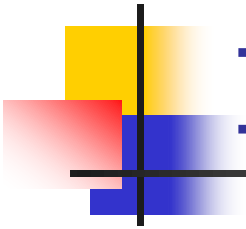
- Robot Interfaces have to be easy to use
  - Robots have to be controllable by untrained users
  - Robots have to be able to interact not only with their owner but also with other people
- Robot interfaces have to be usable at the human's discretion
  - Human-robot interaction occurs on an irregular basis
    - Frequently the robot has to operate autonomously
    - Whenever user input is provided the robot has to react to it
- Interfaces have to be designed human-centric
  - The role of the robot is it to make the human's life easier and more comfortable (it is not just a tech toy)

Adaptation and Learning for Robots in Smart Homes



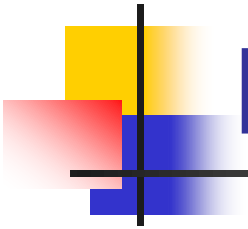
- Intelligent Environments are non-stationary and change frequently, requiring robots to adapt
  - Adaptation to changes in the environment
  - Learning to address changes in inhabitant preferences
- Robots in intelligent environments can frequently not be pre-programmed
  - The environment is unknown
  - The list of tasks that the robot should perform might not be known beforehand
    - No proliferation of robots in the home
  - Different users have different preferences

Adaptation and Learning In Autonomous Robots



- Learning to interpret sensor information
  - Recognizing objects in the environment is difficult
  - Sensors provide prohibitively large amounts of data
  - Programming of all required objects is generally not possible
- Learning new strategies and tasks
  - New tasks have to be learned on-line in the home
  - Different inhabitants require new strategies even for existing tasks
- Adaptation of existing control policies
  - User preferences can change dynamically
  - Changes in the environment have to be reflected

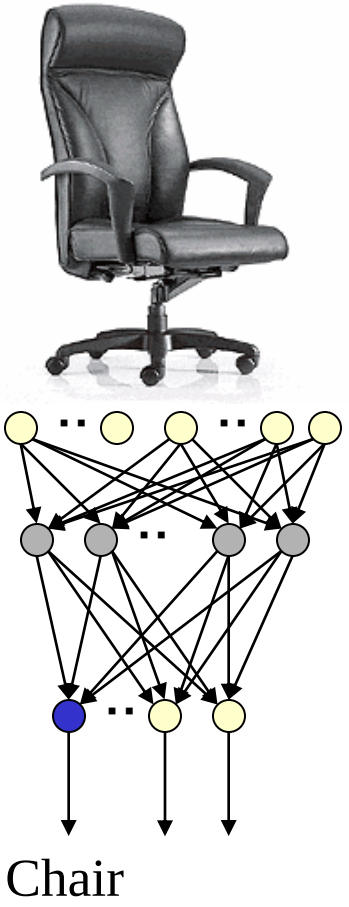
Learning Approaches for Robot Systems



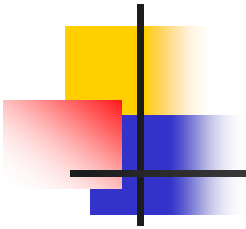
- Supervised learning by teaching
  - Robots can learn from direct feedback from the user that indicates the correct strategy
    - The robot learns the exact strategy provided by the user
- Learning from demonstration (Imitation)
  - Robots learn by observing a human or a robot perform the required task
    - The robot has to be able to "understand" what it observes and map it onto its own capabilities
- Learning by exploration
  - Robots can learn autonomously by trying different actions and observing their results
    - The robot learns a strategy that optimizes reward

Learning Sensory Patterns

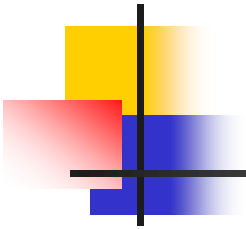
- Learning to Identify Objects
  - How can a particular object be recognized ?
    - Programming recognition strategies is difficult because we do not fully understand how we perform recognition
    - Learning techniques permit the robot system to form its own recognition strategy
  - Supervised learning can be used by giving the robot a set of pictures and the corresponding classification
    - Neural networks
    - Decision trees



Learning Task Strategies by Experimentation



- Autonomous robots have to be able to learn new tasks even without input from the user
  - Learning to perform a task in order to optimize the reward the robot obtains (Reinforcement Learning)
    - Reward has to be provided either by the user or the environment
      - Intermittent user feedback
      - Generic rewards indicating unsafe or inconvenient actions or occurrences
    - The robot has to explore its actions to determine what their effects are
      - Actions change the state of the environment
      - Actions achieve different amounts of reward
    - During learning the robot has to maintain a level of safety



Robotics Laws

- The Laws of Robotics are a set of laws, rules, or principles, which are intended as a fundamental framework to underpin the behavior of robots designed to have a degree of autonomy.

- Robots of this degree of complexity (full autonomy) do not yet exist, but they have been widely anticipated in science fiction, films and are a topic of active research and development in the fields of robotics and artificial intelligence

Isaac Asimov's "Three Laws of Robotics"

- The best known set of laws are Isaac Asimov's "Three Laws of Robotics". The Three Laws are:

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Laws

3 types of Intelligence in Robots

- All artificial AI based systems - real and hypothetical fall into one of three types:

1. **Artificial narrow intelligence** (ANI), which has a narrow range of abilities;
2. **Artificial general intelligence** (AGI), which is on par with human capabilities; or
3. **Artificial superintelligence** (ASI), which is more capable than a human.

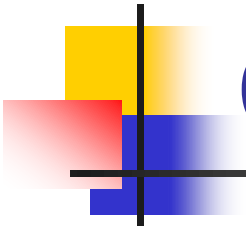
Artificial Superintelligence (ASI)

Artificial super intelligence (ASI), is the hypothetical AI that doesn't just mimic or understand human intelligence and behaviour; ASI is where machines become self-aware and surpass the capacity of human intelligence and ability.

Superintelligence has long been the muse of dystopian science fiction in which robots overrun, overthrow, and/or enslave humanity. The concept of artificial superintelligence sees AI evolve to be so akin to human emotions and experiences, that it doesn't just understand them, it evokes emotions, needs, beliefs and desires of its own.

Artificial Superintelligence (ASI)

- In addition to replicating the multi-faceted intelligence of human beings, ASI would theoretically be exceedingly better at everything we do; math, science, sports, art, medicine, hobbies, emotional relationships, everything. ASI would have a greater memory and a faster ability to process and analyse data and stimuli.
- Consequently, the decision-making and problem solving capabilities of super intelligent beings would be far superior than those of human beings.
- The potential of having such powerful machines at our disposal may seem appealing, but the concept itself has a multitude of unknown consequences. If self-aware super intelligent beings came to be, they would be capable of ideas like self-preservation.
- The impact this will have on humanity, our survival, and our way of life, is pure speculation.



Conclusions

- Robots are an important component in Intelligent Environments
  - Automate devices
  - Provide physical services
- Robot Systems in these environments need particular capabilities
  - Autonomous control systems
  - Simple and natural human-robot interface
  - Adaptive and learning capabilities
  - Robots have to maintain safety during operation
- While a number of techniques to address these requirements exist, no functional, satisfactory solutions have yet been developed
  - Only very simple robots for single tasks in intelligent environments exist

State of the art

- https://www.analyticsinsight.net/the-most-enthralling-robotic-inventions-of-2018/